

**29** Which of the following radioactive samples has the greatest activity?

|          | nuclide                | amount / mole | half-life / day |
|----------|------------------------|---------------|-----------------|
| <b>A</b> | $^{225}_{89}\text{Ac}$ | 0.003         | 10              |
| <b>B</b> | $^{228}_{90}\text{Th}$ | 0.1           | 400             |
| <b>C</b> | $^{228}_{88}\text{Ra}$ | 0.6           | 2100            |
| <b>D</b> | $^{241}_{94}\text{Pu}$ | 1.0           | 4800            |

Answer: A

$$A = \lambda N = \frac{\ln 2}{T_{1/2}} N \quad \text{where } N \text{ is the number of molecules in the sample}$$

So  $N = n N_A$

And the A of each can be compared by finding the nuclide with the greatest ratio of  $\frac{n}{T_{1/2}}$

$$\text{Option A: } \frac{n}{T_{1/2}} = \frac{0.003}{10} = 0.0003$$

$$\text{Option B: } \frac{n}{T_{1/2}} = \frac{0.1}{400} = 0.00025$$

$$\text{Option C: } \frac{n}{T_{1/2}} = \frac{0.6}{2100} = 0.00029$$

$$\text{Option D: } \frac{n}{T_{1/2}} = \frac{1.0}{4800} = 0.00021$$

**30** A radioactive source is placed 1 cm from a Geiger-Muller tube, and various absorbers are placed